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Indicators of transport and accessibility problems in rural Australia

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Abstract

Australia belongs to a small group of countries that are low density but affluent, and characterised by very high vehicle ownership levels but very little public transport in rural areas. It is widely believed that there are few problems of mobility and accessibility, apart from long distances, because everyone has a car. A literature review generally confirms this perception, although there are hardly any suitable local case studies. An analysis of the rural areas of New South Wales, Victoria and South Australia sought *indicators* of transport-related problems, using mainly census data. While the basic urban/rural distinction is clearly defined, *within* the rural areas patterns of vehicle ownership are unusual, and in the remotest areas are not as high as would be expected. Relationships with socio-economic variables are not consistent. Accessibility measures are incorporated in the analysis, and a travel needs index suggested. Appropriate case studies are required to establish travel behaviour patterns and the responses of disadvantaged groups.
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Keywords: Australia; Rural and remote areas; Vehicle ownership; Accessibility; Disadvantaged groups

1. An international context

A recent review of transportation in rural areas has proposed a three-fold categorisation of countries, in which vehicle ownership rates are related to national income levels and population density (Nutley, 1998). While the less developed world—a highly diverse category—is not the concern here, the more advanced countries can be usefully distinguished according to *density*. Those with higher rural densities, as in Western Europe and Japan, have fairly high car ownership levels, but also still have an expectation of adequate public transport to serve the non-car populations, the poor, elderly, young people, and handicapped. In these countries, rural population densities have, historically, been high enough, and distances short enough, to sustain viable public transport services. While these have generally been in decline, under the influence of rising car ownership, national economies are not quite affluent enough to ensure universal access to cars, and hence the need remains for bus and rail alternatives. The basic problem of rural transport—that of dispersed populations attempting to gain access to a necessary range of service outlets located in distant centres, by means of a highly variable distribution of transport resources—is worldwide. Yet it seems that public awareness of such problems, reflected in the academic literature, is greatest in these high density 'Western' countries, and the UK in particular. There is a third category, where perception of the problem is much lower. This comprises those affluent countries that are also characterised by low rural densities and long distances. Here, public transport is not generally viable, and local services have never existed or been long since abandoned. Such countries, especially their rural regions, are dominated overwhelmingly by private road vehicles, to the extent that no problems of mobility or access are believed to exist. This last category comprises only four countries: the USA, Canada, Australia and New Zealand.

This article examines the situation in rural Australia. Within the above framework, Australia would be expected to be analogous to the USA and Canada, with the vast extent of its rural and remote areas and very low densities, posing severe problems, not just for transport, communications and accessibility, but also for service provision and resource utilisation (Lonsdale and Holmes, 1981). A study of the USA, in comparison with the UK (Nutley, 1996), confirms the fundamental differences, not only in objective reality, but also in perception of the 'problem'. An affluent society overcomes low rural densities with saturation coverage of the

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automobile, encouraged by government policies, and long distances are compensated by the relatively low cost of gasoline. In remote areas this is backed up by greater reliance on air transport and telecommunications. What public transport remains is of an inter-city variety, with little awareness of any outstanding needs. Persons or communities with accessibility problems are overlooked, it being assumed that non-car owning populations comprise the elderly and poor only. These perceptions are reflected in the academic literature on transport issues. All initial impressions suggest that rural Australia comfortably fits this stereotype.

Basic census data are sufficient to confirm this (Table 1). National aggregate statistics on car ownership and the journey to work illustrate the very high availability of private motor vehicles and their dominant usage in the travel market. Use of public transport for the journey to work is governed by choice and availability, but it is reasonable to regard the low figures in the table as indicative of its absence or scarcity over wide areas. The Australian pattern seems clearly aligned with that of the USA, and distinct from that of the UK. The multi-car household is commonplace in Australia, as it is in the USA, although not yet a majority. Australia is well on the way to regarding car ownership as an individual attribute, rather than a household asset to be shared.

Definitions of rural areas are problematic in some countries. There is no official definition in Australia, although census data provided by the Australian Bureau of Statistics (ABS) allow the user to construct his own—the definition used in this study will be explained below. Mainly for logistical reasons, so that data can be examined at lower scale-levels, the analysis is confined to south-eastern Australia—the States of South Australia (SA), Victoria, and New South Wales (NSW), including where necessary the Australian Capital Territory (ACT). Data in Table 1 for these rural areas show the expected higher car ownership rates, rising to 55% of households with two or more vehicles, and well under two persons per vehicle. The rural travel to work figures, however, are unusual, with car dependence apparently lower. Use of public transport, as expected, is much reduced, although it is interesting that most of this is by train in NSW and Victoria, but by bus in SA. Most notable are the high figures for walking and 'no journey', which must represent those people working on their own properties in agriculture or other resource-based industries, or people walking to service jobs in small towns.

The objective of the rest of this paper is to search for evidence of 'problems' relating to inaccessibility and lack of transport resources among the population of rural Australia. Particular attention is given to the distribution and character of non-car populations, and the behavioural response to remoteness and long distances. The basic issue is people's ability to make routine journeys within their local area, as an essential component of a normal standard of living. Very high car ownership is unlikely to secure this for everyone, but might cause disadvantaged groups to be neglected. The search begins with a literature review, which aims to reveal the perceptions and 'visibility' of the issue from the amount of work done and perspectives taken. Following this is an analysis from official statistics, primarily the Australian census of 1996.

2. Literature review

As expected, the level of interest shown in rural accessibility and transport problems, as expressed in the Australian literature, is much lower than that in the UK and more akin to that in the USA (Nutley, 1996). This is

Table 1
Basic comparative data on car ownership and travel to work

	Australia (1996)	Australia[a] rural SE States	USA (1990)	USA rural	UK (1991)	UK rural[b] (estimates)
% households with:						
No car	12.3	6.7	11.5	5.9	33.4	20–32
One car only	42.1	38.0	33.8	26.8	43.5	40–50
2 + cars	45.6	55.3	54.7	67.3	23.1	20–35
(of which) 3 + cars	11.7	16.7	17.3	25.1	4.0	3–8
Persons/car	1.96	1.73	1.57	1.43	2.65	2.3–2.8
Cars/household	1.48	1.71	1.67	1.97	0.94	0.9–1.1
% travel to work:						
Car driver	66.4	61.8	73.1[c]	75.8[c]	54.1	45–60
Car passenger	8.0	6.6	13.4[d]	14.7[d]	8.0	7–12
Public transport	9.9	2.1	5.3	0.6	15.8	2–8
Walk	5.8	8.3	3.9	3.0	12.1	10–20
No journey	6.4	16.6	3.0	4.8	5.1	5–15

[a] See text for definition of 'rural'.
[b] There is no official definition of 'rural' in the UK.
[c] Solo car drivers only.
[d] Passengers and drivers sharing.

taken to reflect a low public perception of the issue, typical of affluent, low density countries with very high vehicle ownership. To ensure comparability, the following review is confined to formally published academic/scientific material, and ignores popular media items and 'semi-published' technical reports. The overall impression is of clusters of research interest on cognate topics, which have a marginal bearing on the central problem defined here. A distinct gap is revealed.

2.1. Transport geography

Contributions by geographers to the study of Australian transport are usually national or regional in scale, often including some historical development, and are likely to cover urban transport issues in the major cities, the only domain where a local scale is invoked (e.g. Rimmer, 1987). Considering transport beyond the metropolis, Witherby (1993) briefly summarises the rural situation as overwhelmingly car-dependent, with public transport virtually non-existent except for buses in medium-sized towns, taxis and school buses. There is an important distinction between local travel for basic goods and services, social contact and recreation, and the (apparently increasing) demand for higher-order services which requires very long journeys. Acknowledged losers are "the young, poor, handicapped and frail aged". However, this is presented as only one component of a wider agenda including road and rail infrastructure, freight, and policy issues. Another example is the review of Australian air transport by Quinlan (1998), which focuses on changes in corporate structures. While an emphasis on inter-city routes is easily justified, there is little mention of the specialist and idiosyncratic roles of small scale air services in remote areas; the number of airports with scheduled internal services has declined by nearly half since the 1960s, mainly affecting outback stations.

Local, rural case studies of passenger transport are extremely rare. Perhaps the only study to be fully reported is that of the closure of a branch rail line in New England, NSW, and its replacement by a substitute coach service (Raimond and Parolin, 1992; Parolin, 1996). The policy context was one of public transport rationalisation associated with partial deregulation, gaining pace towards the late 1980s. Coach substitution in this case proved very popular, with a higher standard of service, and also increased business activities in local towns. The curtailment of grain shipments by rail on the same line had little effect on local farming and business (Parolin et al., 1993). In the absence of comparable studies elsewhere, one must be sceptical that these results are representative. There is an urgent need for case studies of travel patterns and accessibility in typical localities with less, or no, public transport provision.

2.2. Transport policy and statistics

One would not expect reviews and discussions of transport policy to be anything other than macro-scale. Conducted either at national (Commonwealth) or State level, these concentrate on issues of political control, financing, deregulation and privatisation or franchising (Scrafton and Starkie, 1985; Scrafton, 1997). An example is the National Transport Planning Taskforce (1994), whose 'strategy' was notable for the absence of any spatial dimension. Contributions by geographers to transport policy questions are also generalised to higher scale-levels. It is also notable how problems of distance, isolation and local development become transmuted to matters of cost, such that the main issue for road haulage, for example, is 'cost recovery' for road maintenance by 'user charges' (Pritchard, 1993). There is a great need for the formulation and impact of policy to take account of 'geographical dimensions' (Taylor, 1991), to consider explicitly rural and remote areas and local scales, and to take a consumer viewpoint.

Official statistics also are far too highly aggregated, very rarely referring to areas below State level. Sometimes there is a distinction between State capital cities and 'other areas'. There are visitor surveys, referring to longer journeys and overnight stays, which are mainly tourist and business trips. The only source of data on everyday trip-making in non-urban areas seems to be Adena and Montesin (1988), which, in a large survey of all Australia, includes figures from different settlement types in 'sampled regions'. From Table 2, people in rural and sparsely settled districts do not necessarily make more or fewer trips than people in urban areas, but the distances travelled and times taken are significantly higher. Differences between the rural hinterlands of towns and sparsely settled areas are of interest, suggesting that the latter do not travel such huge aggregate distances as might be expected (other authors have noted a localisation effect), but longer travel times reflect slow, frequently unsealed, roads. In the absence of any other secondary data, there is no alternative to reliance on the transport-related variables in Australia's quinquennial census.

2.3. Modal studies

Modal or sectoral studies are produced by transport managers, economists and development agencies, and a section of this literature does address the impact on rural and remote areas, especially relating to roads. There is wide agreement on the importance of roads for rural Australia (Kneebone and Berry, 1997), and freight transport is heavily emphasised. Much work of this kind is devoted to economic analysis, such as the implications for agriculture of externally determined road user charges (Bureau of Transport and Communications

Table 2
Basic journey characteristics by settlement type 1985–1986

	State capital city	Provincial city	Rural[a] town-urban	Rural[b] town-rural	Sparsely settled	Mean
Average number of trips per day per person (males)	3.61	3.76	3.80	2.58	3.79	3.62
Average distance travelled (km) per day per person (males)	38.9	46.1	35.0	70.3	55.6	39.5
Average time spent travelling (min) per day per person (males)	79.4	83.2	65.8	76.9	88.4	79.6

Source: Adena and Montesin (1988).
[a] Medium-sized town in the interior.
[b] Rural hinterland of same.

Economics, 1992), although some take the viewpoint of rural exporting industries (Barnard and Kelso, 1994). Few are strongly engaged with specific geographical situations—an exception is Brown (1995) on local access roads to Aboriginal communities.

The coverage of railway matters pays even less attention to local rural services. Although rail freight transport is big business, passenger 'country services' have failed to establish a market niche between the speed of the aeroplane, the convenience of the car and the cheapness of the bus, with the clientèle consisting mainly of better-off tourists, leisure travellers and pensioners (Nash, 1985). A recent factual survey of Australian railways (Productivity Commission, 1999) confirms that rail's share of non-urban passenger transport continues to decline. If this report is representative, there seems to be no interest in 'low volume regional networks' apart from freight.

Bus services in rural areas are generally confined to long distance inter-urban routes and local operations within small to medium-sized towns. Despite a serious lack of data, users of coach services seem to come mainly from low income households and from the younger and older age groups (Bureau of Transport Economics, 1985). Even in remote areas, towns above a certain size support internal bus services (e.g. Milne, 1997). Awareness of disadvantaged groups in need of transport alternatives has long existed, as is shown by an early conference on paratransit (Bureau of Transport Economics and Transport SA, 1980). While this was overwhelmingly urban in focus, paratransit services have since expanded, especially in SA (see below).

2.4. Rural transport

The issues in question are often better approached from a 'rural' rather than a 'transport' viewpoint. However, there are very few publications that consider transport from an exclusively rural perspective, and none by geographers. Two conferences—Byrnes (1987) and the Federal Departments (1991)—both equate transport overwhelmingly with freight and primary exporting industries. The agenda is dominated by policy matters, roads expenditure, user charges and fuel prices.

2.5. Rural geography

Rural studies are more likely to use the local scale and to be people-centred; transport issues can be related to other economic and social problems affecting communities. However, the large literature on 'rural and remote' Australia and settlement patterns very rarely considers transport explicitly. Questions of isolation, distance and communications are usually assumed or implicit, as a kind of subtext in discussions of other things. The density transition, from the population centres on the coast, across the dividing ranges, and declining towards the interior, has fascinated geographers. It would normally be assumed that transport difficulties increase with declining density. If these are valid indicators, the ecumene—the continuously settled zone of permanent agriculture—is defined by Holmes (1987) as equivalent to a minimum density of one person per 8 km^2, while the rural/remote boundary is close to the edge of this zone but also at least 300 km from cities (Holmes, 1988). See Fig. 1. Whether these zones generate different types, or even degrees, of transport problems is not clear.

There is a significant literature on the Australian country town, which deals mainly with the volatility of population levels and long term economic decline. Typical of this is Walmsley and Sorensen (1993), who acknowledge 'isolation' but do not interpret this in transport terms. Health care delivery is used as an example, and it is shown that hospital use declines with distance, but service provision is seen as an issue of political economy. The future prospects of country towns are not seen in terms of overcoming distance barriers, but dependent upon commodity prices, the national economy, and their ability to develop alternative economic bases. It is believed, however, that the smallest, remotest places may not survive.

There have been a few surveys of rural living standards, and these suggest that isolation and transport

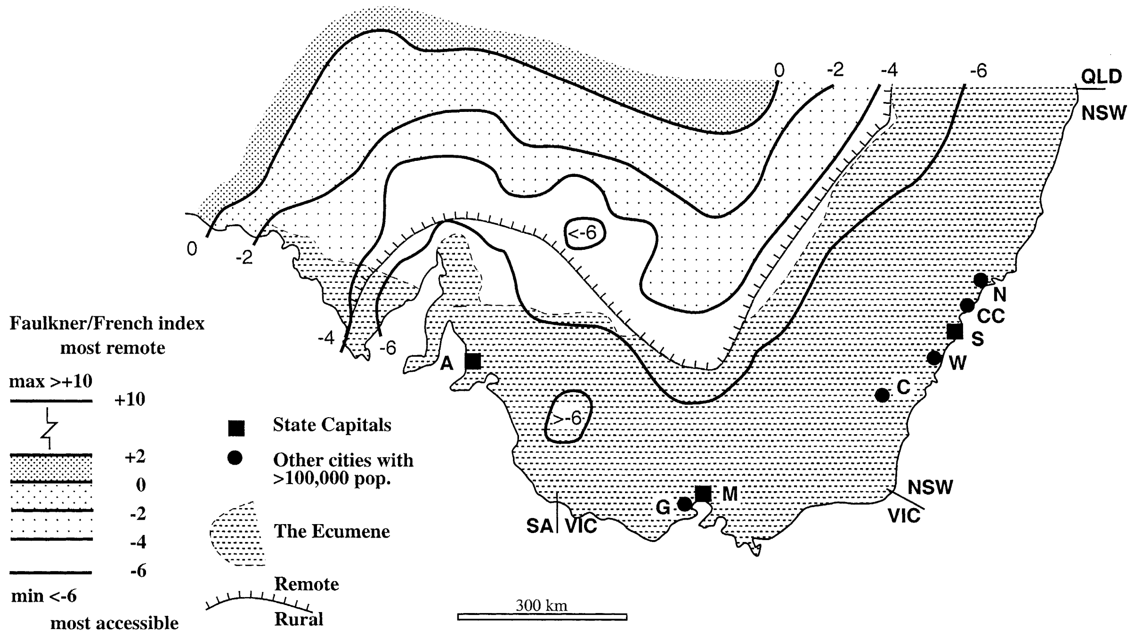

Fig. 1. Aspects of regional accessibility in south-eastern Australia. Key to cities: A, Adelaide; C, Canberra; CC, Central Coast; G, Geelong; M, Melbourne; N, Newcastle; S, Sydney; W, Wollongong; sources: Holmes (1987, 1988).

problems are not necessarily the greatest perceived negative factors. The overall impression is that these are tolerated if the compensations are adequate—incomes, public services, lifestyles—but where these decline, the isolation factor reasserts itself and people leave. A rather dated survey by Stimson (1981) of the Eyre Peninsula (SA) showed that in the context of regional decline, the difficulty and cost of access to a major city (Adelaide) for involuntary trips such as medical treatment heighten the sense of isolation. At the extreme end of the rural spectrum, Holmes (1984) has pointed out the surprisingly high degree of self-sufficiency evolved by the large outback cattle station, through internalising services and reducing outside dependency.

The recent fashion among rural geographers to adopt qualitative or 'post-modern' perspectives has not been helpful in shedding light on these issues, partly because of the high level of abstraction that such approaches involve, but primarily because there has been no attempt, in Australia or internationally, to apply these to questions of accessibility and remoteness. Interest in 'neglected rural others' would be highly relevant to the treatment in Australia of the mobility disadvantaged.

2.6. Rural services

Accessibility in country areas is crucially linked to the location of public and consumer service outlets, and the latter has generated a lot of discussion. Unfortunately, transport is rarely mentioned explicitly, and the implications for access have to be surmised. Typical are the collections of papers by Loveday (1982a,b), which deal mainly with social, political and organisational matters; the problem is small communities rather than long distances. A basic geographical treatment of rural health and education, although dated, is Brownlea and McDonald (1981). Rationalisation of small remote facilities has serious implications. Exceptional methods of overcoming isolation, such as the Flying Doctor Service and distance learning technologies, are duly acknowledged. Experiences of the problems of delivering outback education have been recorded by Fitzpatrick (1983) and Tomlinson (1987). Elsewhere, Wood (1982) has a case study on school transport. Recently added to the agenda is the closure of rural bank branches (Argent and Rolley, 2000). Arguably, improved personal mobility, with people travelling further for goods and services, is a gain in welfare (Rolley and Humphreys, 1993). Much of the debate, however, is of a more theoretical or strategic nature, with services seen as a matter of political economy (Humphreys, 1985, 1993).

2.7. The elderly

The elderly are often singled out in studies of transport and accessibility problems. In the USA, with low rural densities and very high car ownership, it can be shown that non-car households are dominated by the aged, poor, and ethnic minorities. ABS data do not permit simple comparisons but, for the whole of Australia, of the non-car households, 60% were lone persons (many of whom one would expect to be elderly) and 14.5% were one-parent families. Unfortunately, papers on transport for the elderly do not cover rural areas, and papers on the rural elderly (Dempsey, 1981; Williams, 1991) do not cover transport. In the latter, the assumption is that the elderly are totally dependent on the resources of the small towns they live in, and there is no consideration of their ability to travel to other places.

2.8. Rural mobility surveys

Attempting questionnaire surveys in low density areas of Australia poses immense logistical difficulties. A few have been done which report aspects of physical mobility, but as sources of data on travel behaviour they are all in some way incomplete. Focusing on the density gradient from the coast to the interior, Epps (1991) took samples from six towns along a transect through northern NSW. This study concentrated on long distance trips to major cities (Brisbane and Sydney), which were usually multi-purpose, and assumed to be entirely by car. Frequency of trips to State capital cities varied little by distance from the origin point, but remote places generated more trips of equivalent distances to other destinations. Feelings of isolation increased towards the interior, but these were related to the perceived inadequacy of specific services. In the pastoral zone of interior Queensland, an old survey of travel behaviour by Holmes and Brown (1981) showed a very neat inverse correlation between distance from a service centre and frequency of trips to that centre, with the remotest stations compensating with a greater number of local trips.

Stimson's (1981) survey of the Eyre Peninsula, mentioned above, disclosed the remarkable fact that just over 60% of households had travelled to Adelaide by air, only a little less than the proportion that had travelled by car. This region is on the fringes of the State capital's consumer hinterland. All settled rural areas of SA have been covered by a series of postal surveys by Smailes (1995, 1996, 2000). The main focus of these was change in the economic status and viability of country towns, in the context of population change, restructuring, counterurbanisation, and Australia's rural recession (mid-1980s to late 1990s). Comparing surveys done in 1968/1969, 1982/1983 and 1992/1993, there has been a great increase in car-borne mobility and reductions in travel time to Adelaide; shopping and business functions have moved up the urban hierarchy, especially to new suburban and regional centres; local centres still have a role for social activities. Again however, it was assumed that all travel was by car, and it is not known whether costly longer journeys have caused a decline in travel frequency.

2.9. Accessibility

Representations of accessibility in non-urban Australia have all been at the national and regional scales, and have all been of the 'geometric' type based on network analysis, producing isarithmic surfaces. The earliest attempt, by Geissman and Woolmington (1971), covered the south-eastern region between Adelaide and Brisbane, and was based on aggregate distances to all sample points, weighted by population (it is therefore 'population potential' and not 'transport cost'). The resulting quasi-circular pattern of isarithms is predictable given the population density transition. An equivalent but more sophisticated analysis of NSW (Nichols, 1994) was able to express accessibility by travel times and costs, and also to represent access to different destination functions and transport modes.

Some work has specifically focused on 'remoteness' (low levels of accessibility), using similar techniques. The Faulkner/French index, [1] described in Holmes (1988), is a composite measure of standardised scores of distances from sample points to towns of six size thresholds, ranging from 500,000 population down to 500. The resulting map shows a peak of remoteness (high values) in the eastern part of Western Australia, and relative to this, the lowest values (highest accessibility) in the south-east of the country. These are shown on Fig. 1. Such methods are much enhanced by the use of GIS technology, which permits the measurement of actual road distances. Work commissioned by the Commonwealth Government aimed to quantify 'remoteness' in order to assist resource allocation in rural areas. Road distances from all settled localities to service centres of 250,000, 48,000, 18,000 and 5000 population were standardised and combined, to produce an index scaled from 0 to 12 (Department of Health, 1999). This is the Accessibility/Remoteness Index of Australia (ARIA). Scores for the south-east region are mapped in Fig. 2; the use of point symbols here gives a much more precise representation than generalised isarithmic surfaces.

None of these accessibility studies conveys any indication of people's ability to overcome the distances involved. There is no Australian study that employs the rather different techniques needed to express accessibility at local scales.

2.10. Globalisation

Since the mid-1990s it has been common to refer to the overriding context of the restructuring of industry and the global scale of production. Compliant governments have adopted 'neo-liberal' policies favouring the free market and non-intervention. Examples are given by Robinson et al. (2000). In the transport sector these

[1] Attributed to W. Faulkner and S. French (1983) of the Bureau of Transport Economics, Canberra. Although widely cited, this paper was never published and is unobtainable. Some details can be found in Holmes (1988).

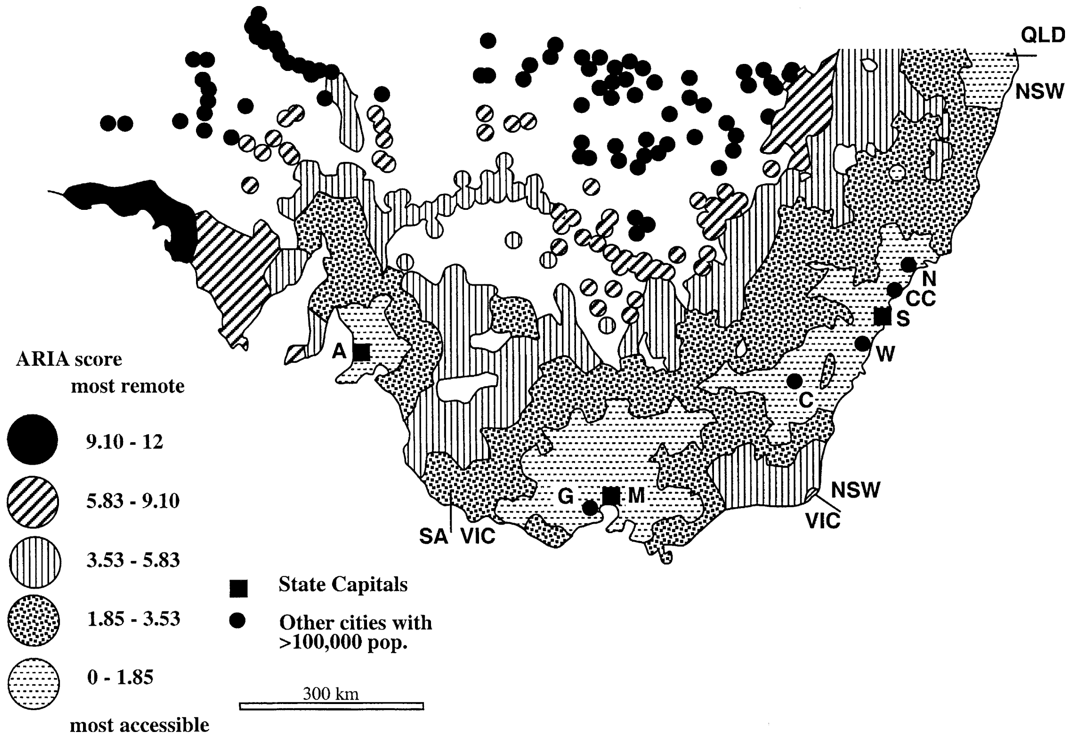

Fig. 2. Accessibility and remoteness by ARIA scores: south-eastern Australia. Key to cities as for Fig. 1; source: Department of Health (1999).

are most clearly seen in air transport and vehicle manufacturing, but more fundamentally,the global organisation of agricultural and mineral industries has caused great uncertainty and stress in the rural producing areas, and has added to existing trends of decline. Some works have attempted to address the local responses to global forces, but these are too highly generalised to make a connection with real-life problems of transport, mobility and access (Lawrence et al., 1999; Pritchard and McManus, 2000; Gray and Lawrence, 2001). In this respect, the main agent is the continuing rationalisation of public and private services, which has severe implications in rural regions for travel needs, accessibility and living standards.

3. Analysis

From the above review there does appear to be a gap in the literature relating to people's ability to make routine local journeys in rural areas. Also, much of the work is rather dated. There is no systematic data source of destinations and distances, journey purposes, travel modes and frequencies; neither is there any consideration of mobility problems of non-car owners or disadvantaged groups. In such low density environments, original, primary data would be obtainable only with great expense and difficulty. The following analysis, therefore, is limited to a search for *indicators* of transport-related problems through relevant variables in the 1996 census.

Having decided to focus on the three south-eastern States, the most appropriate unit areas for initial attention are the Statistical Divisions (SDs)—one level below the State ($N = 31$). As in other national censuses, transport-related variables are limited to vehicle ownership and the journey to work. Socio-economic variables were selected for possible explanatory linkages—population density, age structures and employment, indigenous (Aboriginal) populations, and low income households (defined as less than A$300 per week). Correlation coefficients are only broadly indicative at this scale. Households with no car correlate only weakly with population density ($r = 0.36$, $p < 0.05$) but more strongly with percent population rural (defined below) ($r = -0.68$, $p < 0.01$). Households with three or more vehicles, and 'vehicles per household' have no significant relationship with density, but can be associated with percent population rural (in both cases $r = 0.59$, $p < 0.01$). Further, it is difficult to associate non-car households with disadvantaged groups: the variable might be related to percent indigenous ($r = 0.44$, $p < 0.05$), but there seems to be no verifiable connection with population over 65, low incomes or unemployment. Unlike in the USA (Nutley, 1996), there is no sign that households with 'only' one car have any characteristics of disadvantage.

Journey to work shows some unusual patterns. Public transport is clearly an urban feature, correlating with population density ($r = 0.78$, $p < 0.01$) and negatively with percent population rural ($r = -0.69$, $p < 0.01$), but it is *negatively* related to low incomes ($r = -0.67$, $p < 0.01$). This must indicate where public transport is located, rather than who uses it. Walking to work and 'no journey' are rural characteristics, correlating negatively with density (respectively $r = -0.62$ and -0.54, both $p < 0.01$) and positively with percent rural ($r = 0.60$ and 0.85, both $p < 0.01$), but these are also associated with low incomes ($r = 0.62$ and 0.55, both $p < 0.01$). It must be this feature which, rather oddly, causes 'percent rural' to have negative relationships with travel to work as car drivers and passengers (respectively $r = -0.54$, $p < 0.01$ and $r = -0.40$, $p < 0.05$).

3.1. Definition of 'rural'

Australian census data permits the construction of any definition of 'rural', by selecting urban centres above any chosen threshold population, combining their data values, and subtracting from the SD totals to give the rural balance. Three population thresholds were considered. A limit of 100,000 would exclude the main cities (Figs. 1 and 2). A limit of 20,000 is likely to exclude all towns with internal bus services (e.g. Milne, 1997). There is a consensus among Australian geographers that country towns start to lose their intimate connections with their rural hinterlands somewhere between 5000 and 10,000 population. Also, 5000 is the lowest level of service centre used in the calculation of ARIA scores. An analysis of variance (ANOVA) was conducted on the transport-related variables in the three south-eastern States according to these different urban/rural thresholds. It was found that F-ratios and significance levels consistently increased as the thresholds reduced from 100,000 down to 5000. Rural areas were therefore defined as those remaining after settlements over 5000 population are excluded.

Accordingly, the SDs were divided into urban and rural components. To test the significance of the urban/rural distinction, relative to the possible discriminatory power of any other variable, a series of ANOVAs was conducted on the sample of SDs. The results in Table 3 confirm that, with respect to almost all the selected variables, there is much more variation between urban and rural areas than among the SD geographical units. The unusual relationships with mode of travel to work were noted above. It is surprising that public transport has an insignificant F-ratio: the mean usage is higher in urban than rural areas (3.0% against 1.6%), but when the bigger urban centres are discounted, its usage (reflecting availability) is consistently very low. Among the socio-economic variables, elderly populations are more of an urban characteristic, with rural areas having larger household sizes with more young people, less unemployment and a better dependency ratio. Importantly, neither poverty nor indigenous peoples can be located as predominantly urban or rural.

Table 3
ANOVA by urban/rural areas. South-eastern Australia, 1996

	Dependent variables × urban/rural areas ($n = 61$)		
	F-ratio (d.f. = 1, 59)	Significance	
% households with:			
No vehicle	129.9	($p < 0.01$)	U
One vehicle only	43.1	($p < 0.01$)	U
2 + vehicles	83.2	($p < 0.01$)	R
3 + vehicles	121.8	($p < 0.01$)	R
Vehicles/household	122.0	($p < 0.01$)	R
Persons/vehicle	45.2	($p < 0.01$)	U
Ratio hhs 2 + /1 or 0 vehs	70.7	($p < 0.01$)	R
% no car hhs lone person	2.8	n.s.	
% travel to work:			
Car driver	58.6	($p < 0.01$)	U
Car passenger	82.5	($p < 0.01$)	U
Public transport	2.7	n.s.	
Walk/cycle	6.0	($p < 0.05$)	R
No journey	177.0	($p < 0.01$)	R
Persons/household	29.2	($p < 0.01$)	R
% population > age 65	6.0	($p < 0.05$)	U
% population < age 18	16.6	($p < 0.01$)	R
% pop. Indigenous	0.2	n.s.	
% workforce unemployed	11.1	($p < 0.01$)	U
workers/100 non-workers	5.0	($p < 0.05$)	R
% hhs < A $300/week	0.5	n.s.	

Notes: n.s., not significant; U, variable mean greater in urban areas; R, variable mean greater in rural areas.

The significance of the urban/rural distinction can be checked by using alternative independent (factor) variables. Using households on low incomes, population over 65, percent indigenous, unemployment, and dependency ratio produced very much lower F-ratios with the transport-related dependent variables, although often still significant. The only exception is journey to work by public transport, which is more closely related to low incomes, indigenous peoples, and dependency ratio, than it is to 'urban/rural'. A further check is to use two-way ANOVAs, invoking 'urban/rural' together with 'low incomes', 'indigenous' and 'dependency ratio' as independent variables. In all but one case, most variation is attributable to 'urban/rural', the only exception being 'public transport', which is more closely associated with low incomes. Overall, it seems that there can be no doubt about the strength and validity of the distinction between urban and rural areas.

3.2. Patterns of vehicle ownership

These are the most obvious indicators of transport-related problems, and their spatial expression in rural parts of SDs can be seen in Figs. 3–5. Any preconception that vehicle ownership increases with greater 'rurality', remoteness, or declining density is immediately contradicted. Indeed, there may be grounds for claiming the opposite. The highest car ownership rates are in the rural fringes of Adelaide, Melbourne and Sydney, together with the more settled parts of SA and most of rural Victoria. A better-off commuting element is no doubt a relevant factor. Conversely, it is the remotest

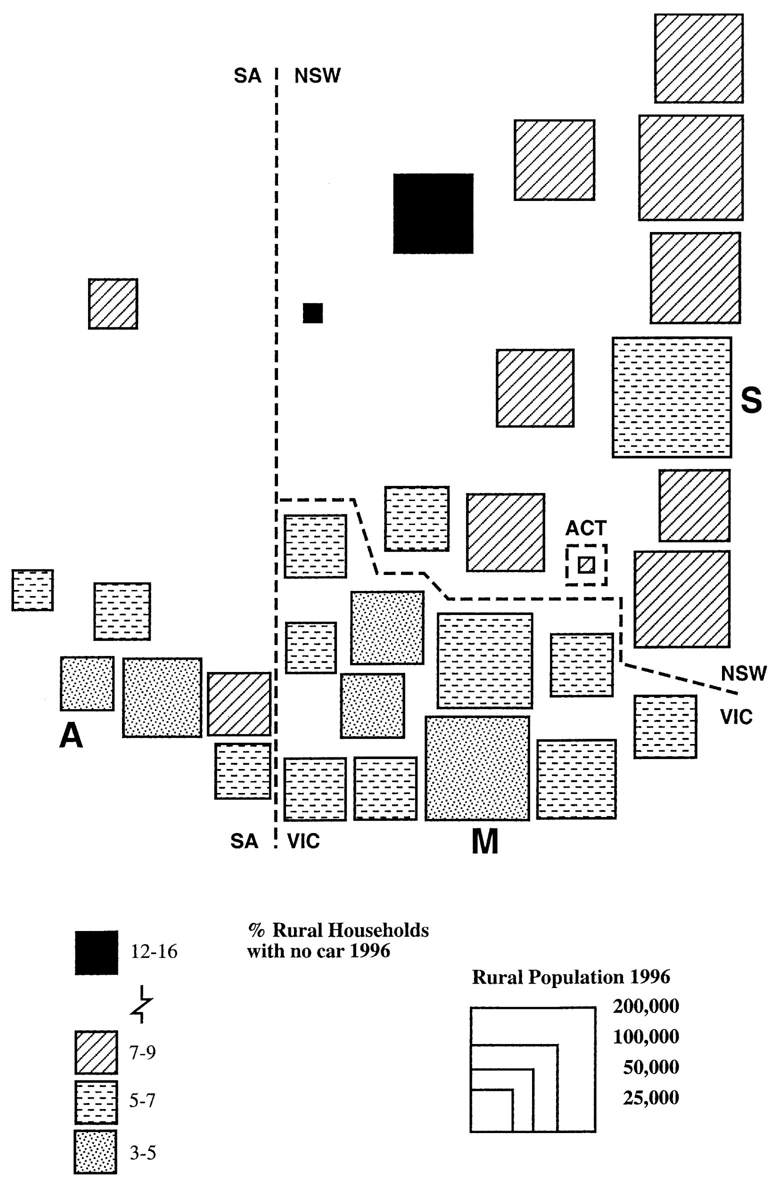

Fig. 3. Rural households with no car: south-eastern Australia. Rural parts of SDs: A, Adelaide; ACT, Australian Capital Territory; M, Melbourne; S, Sydney.

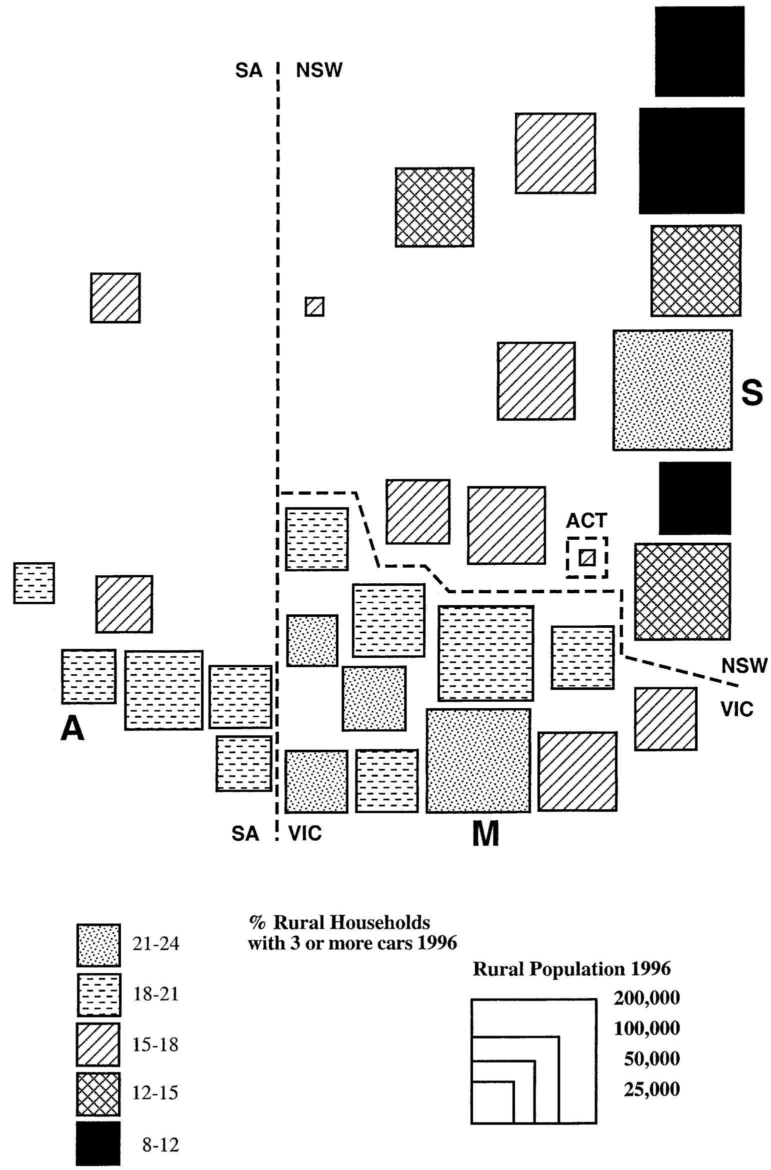

Fig. 4. Rural households with three or more cars: south-eastern Australia (key as for Fig. 3).

areas which have the greatest mobility disadvantage, with non-car households most prominent in the north West and Far West of NSW, and also relatively high in other interior parts of NSW and Northern SA. Comparing with other variables, the only one consistently with above average scores in these interior areas is 'percent indigenous', although this is rarely high enough to be more than a partial explanation.

Among the coastal divisions of NSW the very mixed pattern suggests different degrees of urban influence and widely varying economic fortunes of key centres. This is also indicated by the very great variations in multi-car ownership. Here, the SDs with relatively higher 'no car' rates have no consistent relationship with other variables, but most areas have various combinations of more population over 65, higher unemployment, weak dependency ratio and low incomes. Fig. 5 seems to show a lesser amplitude of persons per vehicle ratios, suggesting that variations in vehicle availability to households are partly compensated by variations in household size.

Correlations can now be re-calculated on the basis of rural areas only. Note that ARIA scores can be included if they are grossed up from a local authority to a rural SD basis. There is more evidence for some of the perverse relationships noted above. *Within* the rural areas, households with no car correlate *negatively* with density and *positively* with remoteness measured by the ARIA score (respectively $r = -0.48$ and 0.66, both $p < 0.01$). Within rural areas, declining density and greater remoteness are associated with less car commuting and more walking or 'no journey' to work. Moreover, within

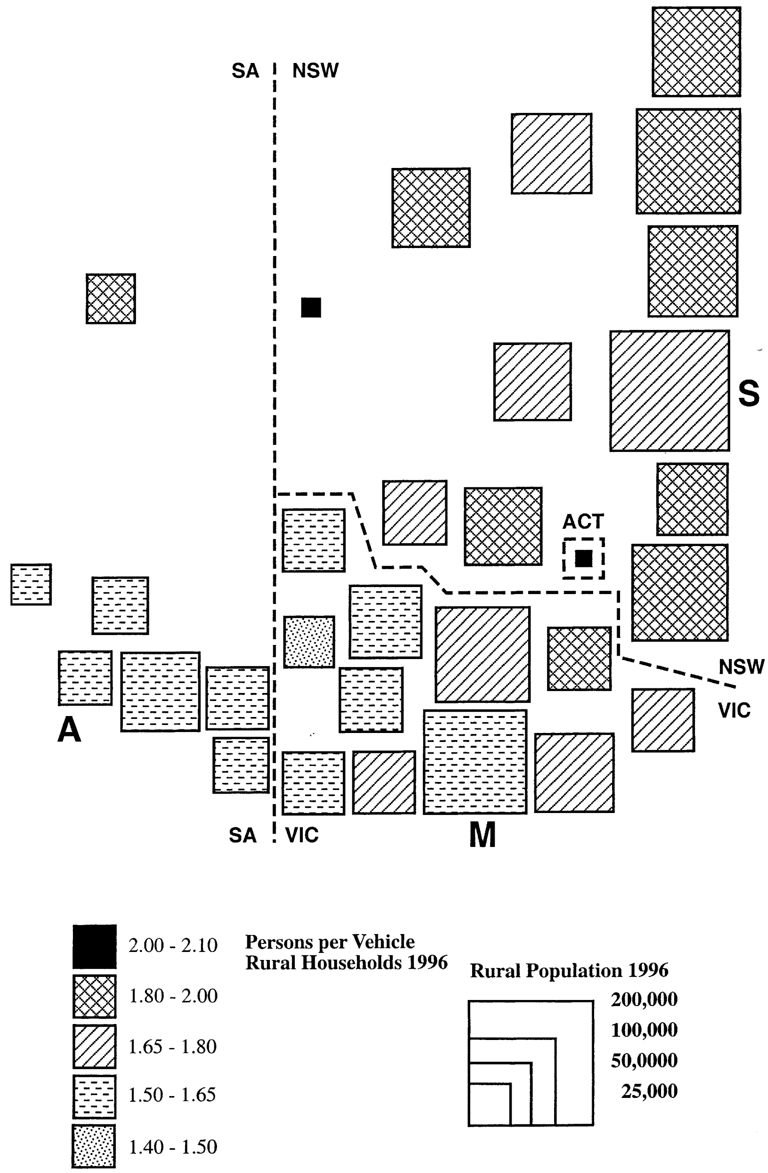

Fig. 5. Persons per vehicle, rural households: south-eastern Australia (key as for Fig. 3).

rural areas there is more significance attached to indigenous populations, unemployment and low incomes. Percent population indigenous relates to 'no car' ($r = 0.86$, $p < 0.01$), persons per vehicle ($r = 0.53$, $p < 0.01$), and journey to work on foot ($r = 0.74$, $p < 0.01$). Percent unemployed is inversely related to vehicles per household ($r = -0.65$, $p < 0.01$). Low income households can now be related to 'no car' ($r = 0.40$, $p < 0.05$), and 'no journey' to work ($r = 0.56$, $p < 0.01$).

3.3. A smaller scale, and the elderly

While the scale and resolution of this study does not normally permit analysis below SD level, there is a case for considering briefly car ownership rates at a more local scale. In order to identify more precise locations of relatively low car ownership, it was decided to examine the Statistical Local Areas (SLAs), which are three levels below the State, and equivalent to local authorities. Rural SLAs—those with no town over 5000 population—that also have over 10% of households without a car, are plotted on Fig. 6. This distribution is very striking, but enigmatic. While there are a few such areas, widely spread, in SA, and only one in Victoria, the great majority are in NSW, in two distinct clusters. The biggest covers most of the north of the State, from New England through the North Central Plains to the Upper Darling and Far West. The other area is a discontinuous band along the Murrumbidgee River. There is no clear relationship with accessibility or remoteness—ARIA scores vary from 2.0 to 10.8, out of a possible range of

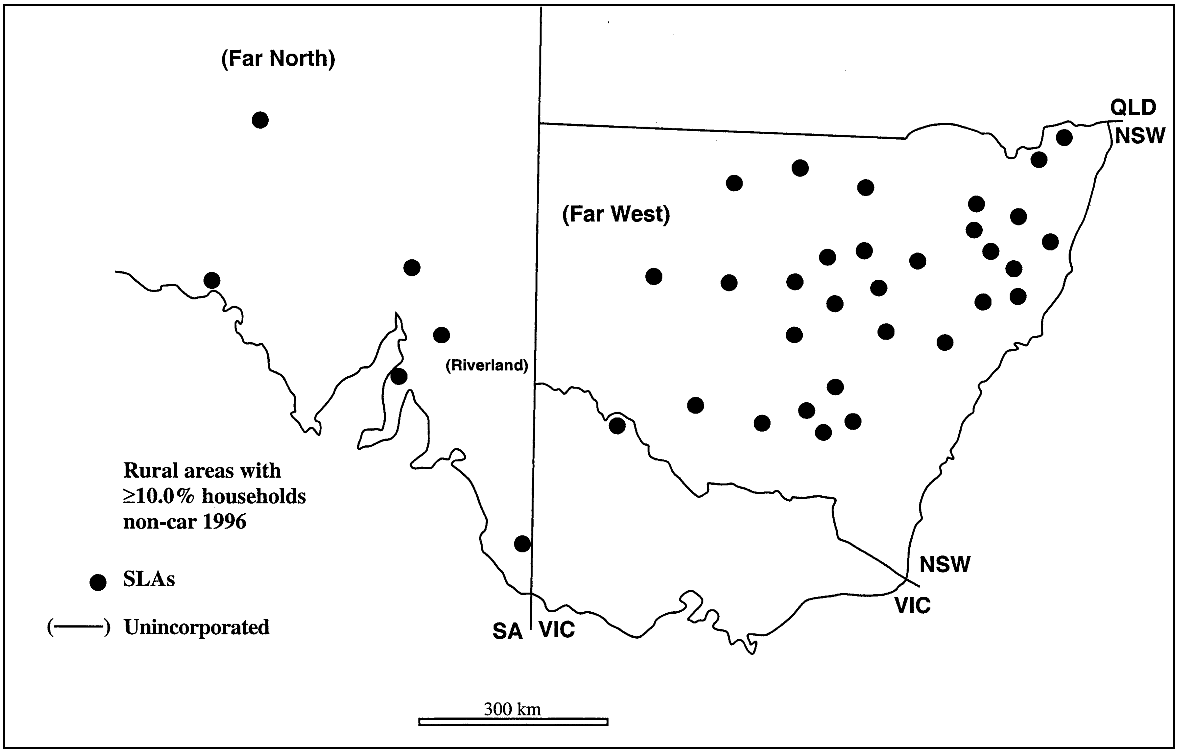

Fig. 6. Rural areas with over 10% households non-car: south-eastern Australia. SLAs not containing towns >5000 population.

0–12. Attempting to associate these 'low car' SLAs with other variables shows that all but one of the 40 areas demonstrate high values (upper quartile) of elderly people, and/or indigenous people, and/or low income households. However, this is unlikely to comprise a sufficient explanation for the extraordinary pattern on the map.

A smaller scale of analysis is also appropriate for further consideration of elderly populations. Accumulated evidence from international studies places great weight on the mobility problems of elderly people in rural areas, but results thus far from rural Australia have given no cause to highlight this factor. This is probably because the population over 65 is fairly evenly distributed, and is not in itself a 'problem' unless mediated by other factors such as low incomes or low car ownership. Published census tables do not relate vehicle ownership to age, but do specify 'lone person households', many of which might be elderly. Amongst almost all of the rural SDs, 58–75% of the non-car households consist of a single person alone. This variable correlates weakly with population over 65 ($r = 0.41$, $p < 0.05$). It also declines with remoteness (correlation with ARIA: $r = -0.65$, $p < 0.01$).

To shed further light on this, customised tabulations were obtained from the ABS which classify vehicle ownership against number of elderly persons in the household. For the whole of Australia, households with at least one elderly person are much more likely to be without a vehicle and much less likely to have more than one vehicle (Table 4). However, by far the biggest anomaly is for households with one elderly person (many of which will be single-person households), while families with two or more elderly persons are not so badly off, and revert more closely to the mean. Samples were taken among the rural SLAs, and some small country towns, in the remoter parts of NSW and SA. First, ANOVAs were done to confirm that variation amongst number of elderly people in the household is significantly greater than variation amongst the geographical areas. Table 4 shows that, while the basic urban/rural distinction in car ownership is maintained, the pattern respecting elderly people is the same, suggesting a 'lone elderly' problem.

3.4. Graphical relationships

Relationships between vehicle ownership, rurality, and other variables may be explored further than is permitted by mere correlations. International studies have shown interesting patterns. UK surveys have shown car ownership increasing down the settlement size hierarchy, through small towns to villages to isolated rural dwellings. At an intermediate scale of unit areas, such as counties, this is not so evident, suggesting instead the dual influence of rurality and income, as is suggested in Fig. 7(a). In the rural USA, even at the crude level of the State, car ownership can be located

Table 4
Elderly persons and car ownership, 1996

No. persons in household > age 65	% households with:			
	No car	One car	Two cars	Three or more cars
Australia				
None	8.9	39.4	38.3	13.4
One	31.4	46.4	16.3	6.0
Two	10.8	61.3	22.8	5.1
Three or more	10.9	49.4	29.4	10.4
One or more	24.4	51.4	18.5	5.7
% no car hhs with one or more persons > 65: ...43.9				
Rural SLAs, NSW + SA (N = 28)				
None	8.6	36.6	37.8	17.0
One	19.8	50.3	19.9	10.1
Two or more	3.7	47.3	35.5	13.5
% no car hhs with one or more persons > 65: ...32.7				
Small urban centres, NSW + SA (N = 16)				
None	13.4	44.3	33.4	9.0
One	35.4	48.1	12.8	3.6
Two or more	10.4	61.0	23.8	4.9
% no car hhs with one or more persons > 65: ...34.1				

according to a poverty factor, a remoteness factor, and a commuter zone (Nutley, 1996, 1998). Rural Australia, however, defies any simple categorisation.

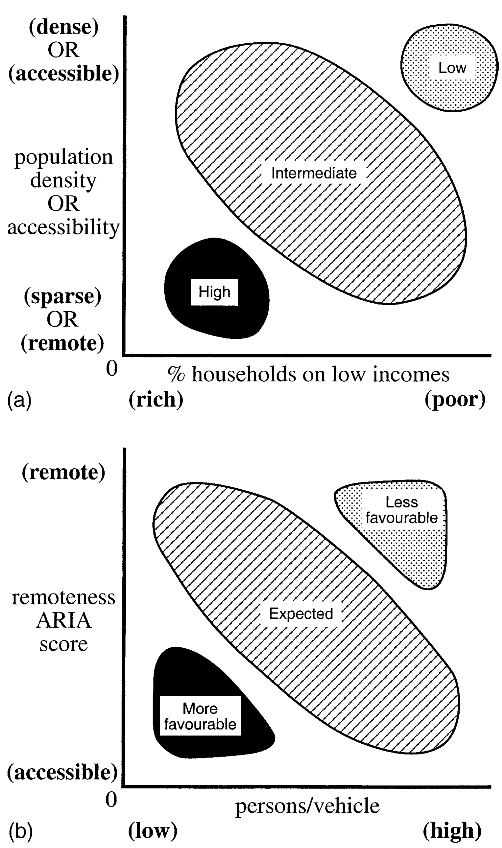

Fig. 7. (a) Top: hypothetical plot of vehicle ownership against density and income, rural areas. (b) Bottom: hypothetical plot of economic and demographic variables against vehicle ownership and remoteness, rural areas.

Attempting to replicate Fig. 7(a) with the 31 SDs produced only a tight cluster of points in the low to medium density range. Using the ARIA score instead of density was slightly more discriminating, showing a few of the remoter locations with above average persons per vehicle, but none showing any discernible relationship with low incomes. Plotting vehicle ownership (as 'persons per vehicle') against the ARIA score allows the data points to be related to other socio-economic variables in addition to low incomes. This might give the pattern hypothesised in Fig. 7(b). Outliers might relate to economic or demographic variables more or less favourable to car ownership. The plotted distribution is shown in Fig. 8. Data points are categorised by the proportion of households on low incomes and by the dependency ratio, which represents the young and elderly populations as well as those employed and unemployed; those with high (upper quartile) indigenous populations are also marked. This pattern bears only a faint resemblance to that suggested in Fig. 7(b), and it might be open to other interpretations. There is some sign of an outlier in the top-right with less favourable characteristics, but hardly conclusive. There is a rather stronger presence in the bottom-left of an outlier with more favourable features. Overall, the pattern is mixed and unpredictable, and in this sense consistent with other results above, and suggests that only a more sophisticated analysis could discern what are the real factors of influence.

3.5. A summary index

Given the lack of case studies in the literature, the analysis has been confined to indicators, often indirect,

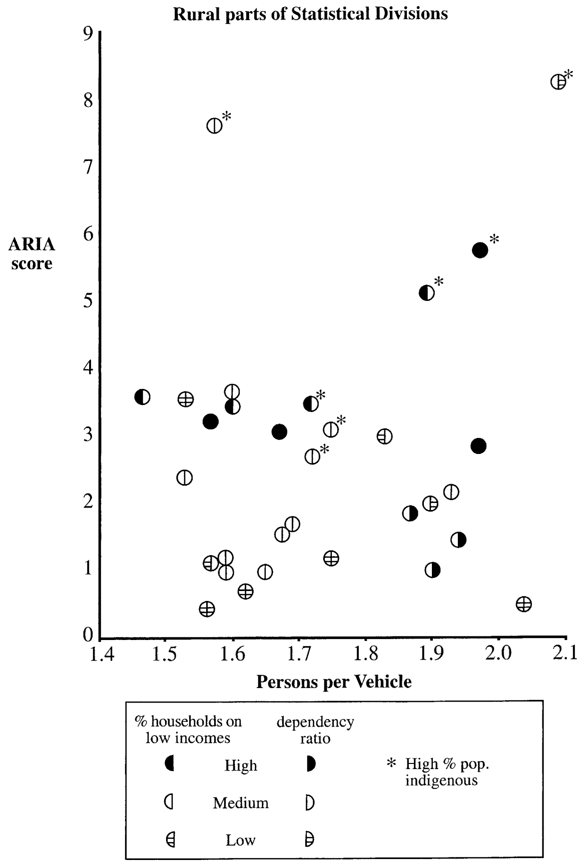

Fig. 8. Economic and demographic variables related to vehicle ownership and remoteness. Rural areas, south-eastern Australia, 1996.

of mobility and transport-related 'problems' amongst the rural population. Some kind of summary index would be useful as an indication of where relative disadvantage might be found, and as a framework for more detailed local investigations. Such an index should be based on comprehensive, publicly available data sources, primarily the census, in the manner of that suggested for the rural USA by Rucker (1984). A 'travel needs index' is proposed as follows:

$$I = (V_0 + (0.5V_1) - T)A/50$$

where V_0 is the percent of households with no vehicle, V_1 is the percent of households with one vehicle, T is the percent of journeys to work by public transport, A is a standardised ARIA score.

V_1 is included to reflect the possibility that not all members of the household have access to the vehicle at all times; the weighting is arbitrary. Journeys to work by public transport are assumed to indicate the general availability of public modes. Lack of transport is obviously compounded by remoteness: the ARIA scores are standardised from their original range of 0–12 to 0–100. The scaling factor of 50 was selected to give the index a maximum value of 100 based on a 'realistic worst case' scenario. The index therefore combines both potential mobility, through transport resources available, and accessibility to service centres. A possibility would be to include demographic variables (as does Rucker's index), such as elderly people and ethnic minorities, but, as these are problematic through lower car ownership, this would have led to double-counting.

Results are mapped in Fig. 9. Lowest apparent needs are in the immediate rural hinterlands of the three State capital cities, and also Canberra, with a fairly regular increase in index values with distance from these, within each State. The northern coast of NSW is coming under the influence of Brisbane. Nowhere in Victoria is remote enough to counteract the high car ownership rates already noted. Conversely, in remote areas where vehicle ownership is not as high as would be expected, the distance factor seems to be dominant, with high values in interior regions and also western SA. The most extreme case—the Far West of NSW—may be anomalous due to its small rural population (outside the city of Broken Hill) and the high (18%) proportion of indigenous people.

4. Conclusions

Although the evidence is rather indirect, there do seem to be fairly clear indications of the likelihood of 'problems' relating to transport, mobility and access to service centres among the populations of rural Australia. While the almost complete absence of public transport outside the towns, and the availability of data, have meant the analysis was strongly concentrated on vehicle ownership, this alone is sufficiently problematic in not being as universally high as generally assumed. From the outset, Australia was placed in the context of other affluent, low density countries where the problems of distance and rural isolation are apparently 'solved' by very high car ownership, perhaps also by low petrol prices, and where the lack of public transport is generally felt to be unimportant. It is assumed that "everyone has a car". The literature review has suggested strongly that this is the general perception: no 'problem' is acknowledged to exist.

While the analysis has shown that it probably does exist, it is certainly not predictable in terms of location, trends, or relationships with socio-economic factors. The finding of relatively lower vehicle ownership in the remotest areas (Figs. 3, 5 and 6) is counter-intuitive and demands further investigation. That there is no consistent relationship with any of the obvious predictor variables—low incomes, unemployment, indigenous people, elderly people—merely indicates the specificity of places, and calls for local case studies to discover precisely who are the mobility deprived, and what is the behavioural response to relative immobility. The index (Fig. 9) might be used as a framework for systematic

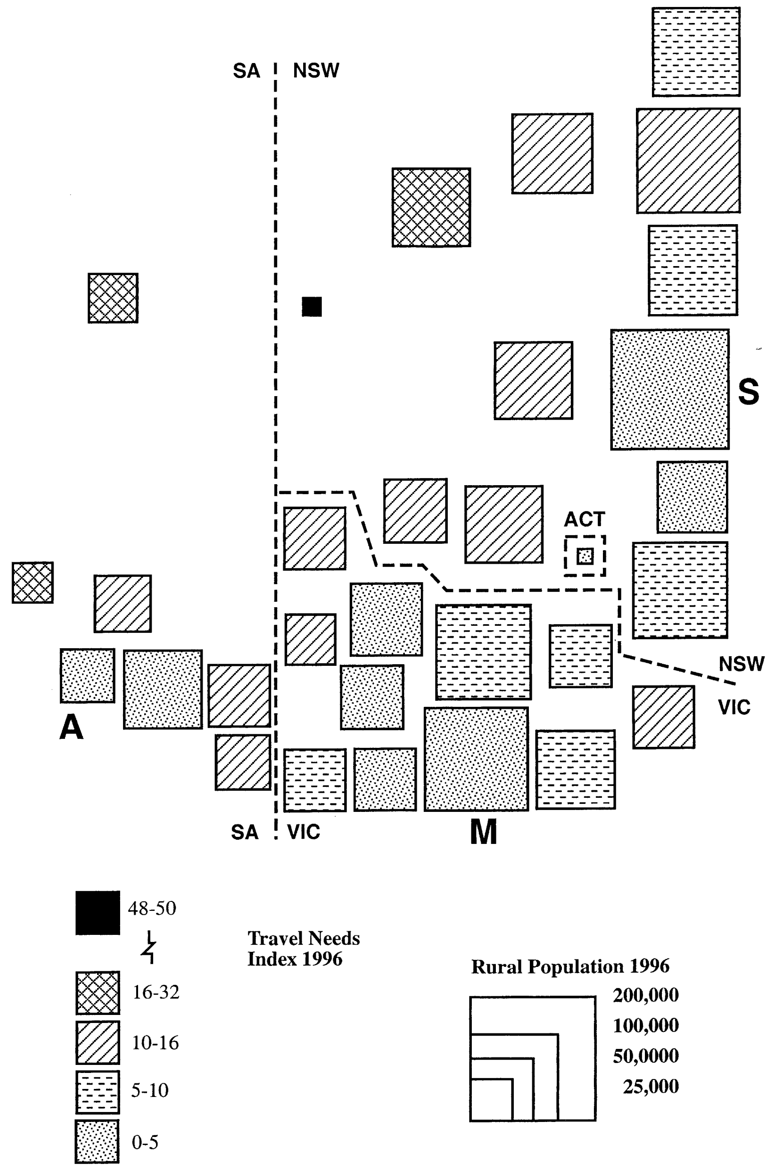

Fig. 9. Results of a proposed travel needs index (see text): south-eastern Australia (key as for Fig. 3).

local studies. It, or an improved version, might be reproduced at SLA or community level so that ARIA scores, which currently represent only road distances, could be fully integrated with measures of transport resources available.

Perhaps the concern should not be focused exclusively on non-car households. In spatially extensive countries with advanced economies, all populations, regardless of private transport resources, seek to mitigate the time and cost penalties of remote living. Two modes of communication are exploited as compensatory expedients. In addition, the 'mobility deprived' in rural locations may have recourse to a third.

(a) *Air transport* is depended upon to a disproportionate extent, in terms of very small settlements serviced and its frequent use for a wide range of journey purposes. But there is a clear need for a study into its precise role in satisfying the travel needs of local communities. Who uses it, and who does not? Does it discriminate? Is it used for relatively lower-order journey purposes such as shopping?

(b) *Telecommunications* are widely available in various forms, and are relied upon to a disproportionate degree in remote areas. Epps (1996) reviews various technologies, prior to the Internet era, with special reference to their social role. But there seems to be no published research on how these technologies are used, apart from just for information, nor on the extent to which they are acceptable substitutes for physical journeys.

(c) *Paratransit* services might be available for disadvantaged groups in local communities, primarily for shorter journeys to essential facilities. Over the last couple of decades there has been much progress in establishing paratransit or community transport schemes. A directory of services in SA shows almost all local authority areas having some such provision (Transport South Australia, 1992). During the 1990s equivalent schemes have developed in much of Victoria and NSW. The SA list reveals that in rural areas community minibuses are most commonly used to transport disadvantaged groups to specific destinations, but some also provide more general-purpose services. While these are welcome safety nets for the mobility deprived, they have not yet attracted any research into their degree of success in meeting needs.

In addition to the specific research suggestions above, there is a clear need for consumer-orientated local case studies as the only means to establish whether there are significant problems of relative immobility and lack of accessibility in rural and remote Australia. It cannot simply be assumed that everyone has a car. It is no surprise that disadvantaged groups in rural areas are politically invisible, but any recognition in policy terms will not come without firm research evidence.

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